Drawn mechanisms: measuring growth in the ability to acquire unfamiliar skills

First draft, 2026-09-22. Author list and venue open. Every number is read from an audited run or from a named analysis file; nothing is quoted from memory.

Abstract

Claims that a system's ability to learn has improved are usually evaluated on fresh instances of a fixed set of task generators. That establishes instance-level transfer and nothing more, and it is compatible with the system having become more familiar with those generators rather than better at learning. We introduce a benchmark in which a task's mechanism, the latent structure that mechanism is drawn from, and that structure's family are all sampled, so instance, mechanism, world and family holdouts are four distinct measurements that cannot be substituted for one another. The hypothesis space is a catalog of 166,596 behaviourally distinct character predicates in a program-synthesis DSL. The truth is always inside it, so correctness is decided by the host interpreter rather than by a maximum over noisy labels, and the resource is an exact count of hypotheses examined.

On this benchmark we confirm, under thresholds fixed before the seeds were drawn and checked by an independent audit, that experience of a latent structure improves the solving of unfamiliar mechanisms drawn from it (+1601 and +1603 tasks of 1,920 over the best experience-free ordering, with a wrong-structure control at baseline), and that experience of 400 other structures improves the acquisition of one never seen (+127 to +204 of 640 at every budget, worth four to five solved mechanisms of the new structure).

We then charge the experience. With every unit of it earned by the agent that benefits from it, a self-improvement loop closes without any outcome feedback: an agent that retains only the atoms of whatever it submitted, right or wrong, raises its own solve rate four- to tenfold at every budget from 128 to 4096, recovering between 3.7 and 68.1 percent of an informed ceiling as the budget rises. We show that the same acquired experience is worth nothing (+86, interval [-40, 222]) or a great deal (+621, interval [384, 841]) depending only on whether it can be applied before the agent's first success, and that reserving half the search budget for an uninformed ordering converts an acquired prior's out-of-support failure into a gain (+121, interval [26, 213]) at no cost in support.

Finally we report four negative results that we did not soften, including a claim withdrawn despite passing its own gate, and three predeclared gates on transfer to a region of the space the training structures never centred on, all of which failed. We give the diagnosis for each and state which of our design rules we expect to survive outside this environment.

1. Introduction

The question we want to answer is whether a system's ability to solve unfamiliar problems and acquire unfamiliar skills has improved, under controlled resources and prior information. The obstacle is not the definition but the measurement. Almost any concrete benchmark fixes a small number of task generators, and a held-out task is then a fresh seed of a familiar generator. Improvement measured that

way is consistent with two very different things: the system has become better at learning, or the system has become more familiar with those particular generators.

The distinction is not academic. In the programme that preceded this work, a verifier trained on more verified labels beat its frozen baseline by +174 on fresh instances of six hand-written task families. Under leave-one-family-out it was worse than the baseline on every one of the six, 716 against 1020, and the regression grew with capacity: at 15, 60 and 120 leaves the in-distribution totals were 1000, 1194 and 1230 while the held-out totals were 762, 716 and 706. The entire measured improvement was familiarity with six generators, and larger models were better at acquiring it. Two further claims in the same programme, both about improvers that select among candidate programs, were withdrawn when a single larger pooled model on the same labels beat them by 35 and 29, and by 50 and 36.

The response to that was not a better improver but a better instrument. This paper describes it and what it has measured so far.

Contributions.

1. A benchmark in which unfamiliarity is graded and drawn rather than labelled. Five splits (new strings, new mechanism, new latent structure, new combination of families, new family) are distinct questions by construction.
2. Two confirmed growth results, with predeclared thresholds, independent replications, structure controls, and an audit that recomputes every number from a decision journal and grades sampled picks through the host interpreter.
3. A measurement of improvement productivity rather than the value of experience: a lifelong setting in which the agent pays for every unit of experience it uses, where the loop closes with no feedback signal at all.
4. A finding we believe generalizes: when an acquired prior can be applied matters more than how good it is.
5. A constructive bound on the failure mode that ended every claim in the previous programme, namely that an acquired prior is a liability outside the support of what it was trained on. Hedging a fixed share of the budget removes the liability and costs nothing in support.
6. Four negative results reported as negatives, including one claim withdrawn after it passed its own gate, and a diagnosis of why a particular holdout could never have been crossed by anything stated over behaviour.

2. Positioning

We are deliberately not writing a related-work section yet. The programme this grew out of adopted a rule that no prior work is cited until it has been read directly, and a number of obvious neighbours, including work on self-improving program synthesis, benchmark design for emergent ability claims, item response models for capability measurement, observational scaling laws, proper scoring of research forecasts, and reward-model overoptimization, have been identified from listings and summaries rather than full texts. Citing them from summaries would be the same error this paper is about.

The reading list is fixed and is a blocking item before submission. We state here only the two positioning claims we are confident of, and invite correction.

First, the benchmark's central move is to make the hypothesis space finite and enumerable, so that a resource budget is an exact count and the oracle is the truth rather than a maximum over observed scores. We are not aware of a learning-ability benchmark that takes that trade, and it is the design decision a reviewer should attack first.

Second, the distinction we draw between the *value of accumulated experience* and *improvement productivity* is, as far as we can tell, usually left implicit. Section 6 shows them coming apart by the whole of an effect.

3. The benchmark

3.1 Mechanisms

A mechanism is a character predicate written in the host interpreter's ordinal expression language, so the interpreter, not the generator, decides what a predicate means. Atoms are depth-one predicates over `ord(c)` drawn from nine families: residues of a modulus, single bits, thresholds, popcounts and parities of masked words, quotient residues, gcd tests, xor-then-modulus and rotate-then-modulus. Enumerating every atom, every cross-family pair under three boolean operators, and every complement, then keeping one representative per distinct truth mask over a 256-character alphabet, yields a catalog of **166,596 behaviourally distinct predicates**, of which 105,126 have density between 0.2 and 0.8 and can therefore generate tasks.

The catalog is a function of the atom list alone. It is identified by a digest recorded in every run that uses it.

3.2 Tasks

A task shows eight examples of one mechanism and hides sixteen. An example is a string of 2 to 5 characters satisfying the predicate followed by 2 to 5 that do not; the answer is the first block. The alphabet is `chr(0x800 + i)` for i in [0, 256), all Unicode category Lo, so the interpreter's built-in letter/digit transition carries no signal and the predicate has to be invented rather than looked up.

A hypothesis is a predicate `X` read as *the answer is the longest prefix of characters satisfying X*. It reproduces an example exactly when every character of the first block satisfies `X` and the first character of the second block does not, which is two integer tests against bitmasks. The rendered program is

```
Concat(Slice({"kind":"start"}, {"kind":"run","predicate":X,"ordinal":1,"side":"end"}))
```

whose interpreter reading is "up to the end of the first run of X". The two agree in one direction only: a hypothesis passing the mask tests always passes the interpreter, while the interpreter additionally accepts programs whose first run begins after position zero. Grading is therefore sound and slightly conservative. Over 31,336 audited decisions there were **zero** cases where grading credited an answer the interpreter rejects, and 169 where the interpreter accepts an answer grading counts as a miss, a rate of 0.54 percent; an exhaustive scan over all arms of 400 tasks put the rate at 0.14 percent (105 of 75,063). The true mechanism always satisfies the prefix reading, so no task is made unsolvable by it.

3.3 Worlds

A **world** is a latent structure a task's mechanism is drawn from. In the main construction a world is 14 atom keys sampled from three atom families, and its members are the catalog rows built only from those keys and lying in the drawable density band, typically about 150 of the 166,596. Nothing in a task reveals the keys. A world with too few members is rejected rather than padded, so world size never silently becomes the variable under study.

A second construction, used in section 7, defines a world behaviourally: a centre atom plus the 14 atoms nearest it under a hidden sparse weighting over descriptor dimensions. Every atom has a descriptor, including atoms no training world contained, so a prior stated over that weighting is defined for them.

3.4 The solver interface, and why ordering is the only variable

An arm orders the catalog, scans the first `B` entries, keeps the candidates consistent with the eight visible examples, and picks among those by one fixed free rule (smallest positive set, ties by lowest complexity). Ordering is therefore the only thing that differs between arms and `B` is the only resource. Two consequences follow. "Give the baseline a larger budget" is always available as a default, and the budget-equivalence of an arm's advantage is directly readable.

All arms share one random tie-break. Breaking ties by catalog order, which is complexity order, gave the free rules a systematic preference among rows they cannot distinguish and made their solve rates jump between budgets for reasons unrelated to their scores.

3.5 Splits

split	what is fresh
instance	the example strings
mechanism	the catalog entry, never solved before, different truth mask
world	the key set
family combination	the three families in combination, each seen separately
family	at least one atom family no training world contained

4. Protocol and statistics

Arms, splits, budgets, statistics and thresholds are fixed in a protocol document written before the seeds are drawn. Confirmatory runs use two replications on independent seeds and pass only if both pass. Comparisons are paired over tasks with 20,000 bootstrap resamples; intervals are 2.5 and 97.5 percentiles of the resampled sum.

Test mechanisms are removed from every training world's solved sample, so no tested mechanism was observed in any world.

Every confirmatory run writes a per-decision journal. An independent audit recomputes every reported rate and every gate from the journal alone, rebuilds the experience-free ordering and repeats the scan on sampled decisions, checks that every recorded pick reproduces the visible examples, and runs sampled picks through `evaluate_program`. The audit's first pass failed, which is how the grading asymmetry in section 3.2 was found.

Runs record wall time, CPU time, peak resident memory, and the count of tasks with no consistent candidate.

4.1 Screens before arms

Two preconditions are screened before any improver is built, following the practice adopted after the previous programme.

Power. Eight examples leave 8.6 consistent candidates on average and a unique one 41.9 percent of the time, and some consistent candidate is correct on every task.

Learnability in selection. Given the whole consistent set at unlimited budget, a gradient-boosted selector over 4,000 drawn mechanisms reaches 0.785 under a mechanism holdout against 0.787 for "pick the smallest consistent set" and 0.788 for "pick the candidate most agreeing with the rest", and 0.825 under a family holdout against 0.830 and 0.825.

rule	mechanism holdout	family holdout
fitted selector, 60 leaves	0.785	0.825
smallest consistent set	**0.787**	**0.830**
largest agreement with the rest	**0.788**	0.825
lowest complexity	0.756	0.787
random consistent candidate	0.651	0.715

Fitting adds nothing over a free rule. **The residual is not in selection.** Everything after this measures reaching a good hypothesis, not choosing among the ones already in hand.

5. Confirmed results

Two replications, 16 worlds per split, 40 test mechanisms per world, 1,920 tasks per replication, no task without a consistent candidate. Cost: 220 s wall, 254 s CPU, 192 MB peak for both replications. The audit passes: every rate and every gate recomputed to the reported value, all 3,840 rebuilt picks matched, no pick inconsistent with its visible examples, no tested mechanism in the experience pool used to choose for it.

5.1 G1: experience of a structure, on unfamiliar mechanisms of that structure

At a budget of 256 of 166,596, over the 1,920 tasks of a replication:

comparison	replication 1	replication 2
experience of 32 of the world's mechanisms minus the best experience-free ordering	**+1601**	**+1603**
minus the same rule fed another world's mechanisms	**+1643**	**+1644**

Solve rates at that budget: 0.034 to 0.048 for catalog order, 0.003 to 0.009 for density order, 0.005 to 0.052 for another world's experience, 0.852 to 0.902 for the world's own. The development curve is

monotone in experience: 0.001, 0.018, 0.057, 0.165, 0.326, 0.644, 0.775, 0.803 at 0, 1, 2, 4, 8, 16, 32 and 64 solved mechanisms.

At 64 solved mechanisms and a budget of 256, experience beats every free ordering *at unlimited budget*, because the ordering also places good hypotheses ahead of bad ones among those it reaches.

We report G1 as the floor and note that it is close to tautological: knowing which 150 of 166,596 hypotheses matter is worth a great deal. Its value is that the wrong-world control makes it a clean instrument check, and that no arm here fits a model. The ordering is a count of atom occurrences, so no gain can be credited to capacity.

5.2 G2: experience of other structures, on a structure never seen

Worlds whose three atom families were each trained on but never in this combination. Budget 1024, tasks out of 640, both replications:

k	minus the best free rule	minus the no-structure control	minus the label-scrambled control
1	+127 [103, 150] / +127 [102, 152]	+110 [86, 134] / +127 [102, 152]	+119 [95, 142] / +128 [103, 153]
2	+190 [163, 217] / +188 [163, 214]	+176 [152, 200] / +185 [161, 210]	+191 [167, 215] / +166 [141, 191]
4	+204 [178, 230] / +201 [174, 227]	+219 [193, 245] / +189 [162, 216]	+214 [189, 239] / +170 [144, 196]

Every interval entirely positive, and the same sign at every budget from 256 to 16,384. The no-structure control is the identical estimator fitted on worlds drawn without family structure; the scrambled control is the same fitted estimator with its atom labels permuted.

Sample efficiency. At budget 1024, one solved mechanism plus experience of 400 other worlds reaches 0.331 and 0.338, which the free rule needs four to five solved mechanisms of the new world to match.

The mechanism is legible. A world takes 14 keys from three families, so atoms of one family co-occur across training worlds; the prior generalizes from an observed atom to atoms that travelled with it. That is why it reaches an unfamiliar combination of families, and why section 7 goes the way it does.

6. Compounding, and the earned loop

The results in this section have controls and intervals but no predeclared gate. We label them development evidence.

6.1 Compounding

G2 is a level: a fixed quantity of prior experience buying a fixed advantage. Compounding is a claim about the derivative. Fitting the same estimator on the first 25, 50, 100, 200 and 400 training worlds, and reading the margin over the free rule on 20 worlds whose family combination no training world used, 800 tasks:

k	budget	25	50	100	200	400	400, unstructured
1	1024	-26	-7	+20	+37	+54 [29, 80]	-4 [-26, 18]
2	1024	+21	+60	+79	+90	+122 [94, 150]	-14 [-37, 8]
4	1024	+20	+47	+55	+94	+104 [77, 131]	-20 [-41, 1]
1	4096	-95	-96	-67	-35	+5 [-19, 29]	-64 [-82, -47]
2	4096	-146	-101	-47	-10	+43 [19, 67]	+10 [-8, 28]
4	4096	-77	-20	+33	+80	+102 [73, 131]	-35 [-64, -7]

Monotone in all six cells, not flattened at 400 worlds. The control sweep on worlds drawn without family structure is flat or negative everywhere and never becomes monotone, so this is structure accumulating rather than an estimator warming up.

Two cautions. This is not recursive: the estimator that turns worlds into a prior is fixed and only its input grows. And the small-sample cells are a warning, not noise: a prior fitted on 25 worlds is worse than no prior, by -95 to -146 with intervals entirely negative.

6.2 The earned loop

Everything above hands an arm its solved mechanisms. Solving them cost budget and none of it was charged, so sections 5 and 6.1 measure the value of accumulated experience and not improvement productivity.

An agent meets 120 tasks from one world, each a mechanism it has never seen, spends the same scan budget on every one, and starts knowing nothing. Whatever it learns comes from tasks it has already paid for, so total resource is identical across arms. The task stream is shared, so every comparison is paired on the same mechanisms in the same order. 24 worlds.

arm	B=256	first 10 to last 10	B=1024	first 10 to last 10
never updates	0.8	0.008 to 0.013	1.8	0.021 to 0.013
keeps the atoms of whatever it submitted	3.5	0.013 to 0.037	**19.7**	0.037 to **0.292**
keeps them only when told it was right	2.4	0.013 to 0.017	17.2	0.042 to 0.287
told the true mechanism every time	95.6	0.250 to 0.883	99.0	0.300 to 0.925

Against the arm that never updates: +429 [289, 576] and +371 [255, 493] at budget 1024; +66 [28, 112] and +40 [12, 72] at 256.

The loop closes, and it closes without feedback. The blind arm consults nothing outside the task. Every candidate it submits was consistent with eight examples, which is apparently enough for its atoms to be worth keeping even when the answer was wrong, and it is at least as good as the arm told whether it was right. One bit of outcome feedback per task buys nothing here.

There is no takeoff threshold. Six budgets, 16 worlds of 100 tasks:

budget	never updates	blind	informed ceiling	blind, first 20 to last 20	blind minus static
128	0.2	1.8	48.5	0.003 to 0.031	+25 [4, 51]
256	0.5	4.9	78.5	0.013 to 0.091	+71 [19, 144]
512	0.7	8.7	82.2	0.022 to 0.172	+128 [69, 199]
1024	1.3	13.3	80.6	0.044 to 0.219	+192 [118, 280]
2048	1.8	28.2	79.4	0.084 to 0.447	+422 [299, 544]
4096	3.4	53.2	78.1	0.200 to 0.772	+796 [730, 858]

The loop closes at every budget with an interval entirely positive, and the last-twenty rate is 4 to 10 times the first-twenty rate throughout. What grows with the budget is the share of the informed ceiling recovered: 3.7, 6.2, 10.6, 16.5, 35.5 and 68.1 percent. The ceiling saturates by budget 256, because an agent told the true mechanism knows its world's 14 atoms within a few tasks. The earned agent is limited by the rate at which correct atoms arrive, and the budget only sets that rate. An earlier two-point version of this measurement suggested a threshold; six points show a smooth curve and none.

6.3 When a prior applies matters more than how good it is

The prior confirmed in G2 does not accelerate the earned loop: +86 [-40, 222] at budget 1024 and +34 [-1, 83] at 256, both crossing zero. The reason is structural rather than statistical. That prior conditions on the atoms of mechanisms the agent has already solved, and an agent that has solved nothing has nothing to condition on, so for exactly the stretch that decides whether the loop starts it is the no-experience ordering.

What the training worlds also know, and that prior never used, is which pairs of atoms belong together in any world at all. That is unconditional and available before the agent sees a single task. Starting the earned agent from that ordering and changing nothing else, tasks solved of 100, 16 worlds:

group	budget	static	earned loop	cold-start prior	permuted control	ceiling
families training held	256	0.6	3.4	**23.0**	3.9	84.6
families training held	1024	1.9	13.6	**52.4**	17.6	83.4
a family it did not	256	0.3	2.3	7.2	5.4	81.2
a family it did not	1024	1.3	13.7	10.7	**19.9**	81.4

On familiar families: **+314 [193, 432]** and **+621 [384, 841]** over the earned-loop champion, with the permuted control beaten by +306 [212, 396] and +558 [350, 763]. The share of the informed ceiling recovered goes from 4 to 27 percent at budget 256 and from 16 to 63 percent at 1024. Most of that is knowing which atoms appear in worlds at all rather than which pairs go together: replacing pair co-occurrence with each atom's own rate loses only +56 [17, 102] at 256 and +53 [-5, 106] at 1024.

Outside the trained family support the same prior is worse than its permuted control, -147 [-240, -60] at budget 1024.

6.4 Hedging bounds the loss

Every acquired prior in this work fails in one place and no other. Two repairs are available. **Hedge**: reserve a fixed share of the scan budget for the ordering the agent would use with no prior, which needs

nothing to be detectable because the loss is bounded by the reserved share whatever the prior believes. **Detect**: switch the prior off when the fraction of the agent's own solved atoms appearing in no training world exceeds the share a hedge would have reserved.

group	budget	prior	hedge 25%	hedge 50%	detector
familiar families	256	+274 [176, 412]	+263 [170, 389]	+235 [164, 327]	+274 [176, 412]
familiar families	1024	+298 [134, 507]	+301 [158, 496]	**+325 [186, 515]**	+298 [134, 507]
unfamiliar family	256	+95 [46, 146]	+98 [50, 147]	+87 [46, 130]	+95 [46, 146]
unfamiliar family	1024	**-5 [-93, 76]**	+65 [-32, 165]	**+121 [26, 213]**	-5 [-93, 76]

Reserving half the budget converts the out-of-support failure into a gain and costs nothing in support. The detector never fired in any group at any budget: foreign atoms imported from wrong answers come mostly from the families training did contain, so the statistic is diluted. The hedge needs no statistic.

7. Negative results and the support boundary

7.1 A claim withdrawn despite passing its gate

G3 asked whether an acquired prior transfers to a world containing a family no training world held. Its gate read budget 1024, at k of 2 and 4, and it passed: +35 [12, 58] and +22 [2, 42] in one replication, +33 [13, 53] and +23 [4, 42] in the other, with the permuted control beaten in all four cells.

The same comparison at other budgets, same runs:

budget	k=1	k=2	k=4	k=8
256	+13 / +13	+29 / +12	**-16 / -17**	**-18 / -52**
1024	+7 / -5	+35 / +33	+22 / +23	+22 / +7
4096	**-46 / -63**	**-41 / -39**	-10 / +3	+21 / +42
16384	+46 / +57	+29 / -1	-17 / -8	-26 / -9

Bold cells have intervals entirely negative. A gain confined to one budget with significant losses on either side is a budget-specific artifact. Reading the gate at a single budget was selection even though it was declared in advance. We withdrew G3 and made "read a gate across the whole resource range" a standing rule.

7.2 Why that boundary could never have been crossed

Four attempts to transfer to a world built from a family no training world contained failed. A fifth measurement explains all four.

Clustering the 368 atoms by the whitened Fourier and Walsh magnitude spectra of their truth masks and asking how well the clusters recover the nine families gives, at the best k over 6 to 32, **adjusted Rand 0.144**. Per family: `threshold` is recovered perfectly, `parity` 0.86, `bit` and `rotmod` 0.75, and the two normally withheld families are captured at 0.69 and 0.47 with 0.68 and 0.72 of their home clusters foreign.

This is not a weak descriptor so much as a category that is not behavioural. In these worlds a family is the *name of an arithmetic operation*: `mod:2:0` and `mod:16:7` share a label and nothing observable. The family holdout asked for generalization across a boundary nothing computable on an atom can see, and no representation defined on behaviour could have crossed it. The repeated failure is a fact about how the worlds were built, not a law about transfer, and it retro-explains the verifier collapse described in section 1.

7.3 Three predeclared gates on a boundary that is crossable in principle

We rebuilt worlds behaviourally: a hidden sparse weighting over descriptor dimensions, drawn once per replication and shown to nobody, with a world a centre atom plus the 14 atoms nearest it under that weighting. Every atom has a descriptor, so a prior stated over the weighting is defined for atoms no training world contained.

gate	what it asked	outcome
G4	a fitted metric beats the free unweighted rule *and* a dimension-permuted copy of itself, at every budget	**failed** on the control clause: beat the free rule in 15 of 16 cells, indistinguishable from the permuted copy at budgets 256 and 16,384
G5	the same with the target made identifiable	**failed** one cell of sixteen in each replication, k=2 at budget 256
G6	the same with doubled power at fixed effect size	**failed**: replication 1 passed every cell, replication 2 failed three

Between G4 and G5, two screens found that G4 had asked for something not identifiable. Fitting to squared differences rather than elementwise products, which is the form a weighted distance actually takes, moved recovery of the true active dimensions from 0.28 to 0.29, so the estimator was not fitted to the wrong thing. The catalog has 368 atoms and the descriptor 385 dimensions, so a weighting over those dimensions is not determined by the distances among those points:

descriptor width	8	16	32	64	128	385
active dimensions recalled	1.00	1.00	**0.92**	0.90	0.70	**0.33**
correlation with the truth	0.84	0.92	0.90	0.86	0.85	**0.13**

G5 repeated the question at width 32 with 8 active dimensions, both fixed in the protocol before any task was run at that width. The weighting was recovered at Pearson 0.939 and 0.976 and the metric beat both comparisons in 15 of 16 cells in each replication. The one cell it failed, k=2 at budget 256, is where the protocol asked for something close to unattainable: in one replication the *true* weighting beats the free rule there by only +19 [3, 35], so a floor of +15 demanded four fifths of all the headroom in the cell. A floor stated as an absolute count and applied uniformly to cells with very different headroom is a protocol flaw of the same species as G3's, and we record it as one.

G6 added power at a fixed effect size and nothing else: 32 worlds per split, 1,280 tasks, floor scaled from +15 of 640 to +30 of 1,280, fresh seeds. Replication 1 passed every cell. Replication 2 failed three, and not only for want of power: at k=2 and budget 256 the permuted copy beat the metric itself, -22 [-36, -9].

No claim is made. Every cell at budgets 4096 and 16,384, in every replication of G5 and G6, is positive against both comparisons with intervals entirely positive, some by +196 to +224. That is a

description, not a claim. A fourth gate restricted to those budgets would be a range chosen from results already seen, which is what withdrew G3, so we stopped. What the environment has established is that the headroom is real, that an identifiable target is recovered almost exactly, and that converting recovery into a reliable margin at every resource level is not something a diagonal metric over these descriptors does.

7.4 Two estimator upgrades that looked right and were not

Replacing pairwise-maximum affinity with joint retrieval over training worlds, which is the better-specified model of the generative process, loses by -22 to -54 of 640 across the seen-combination and new-combination splits, with intervals entirely negative in most cells. Weighting concentrates the estimate on the handful of training worlds containing the one to four observed atoms, while a maximum over pairwise counts pools the same evidence across every world containing that atom. At the evidence levels where a prior is worth having, the cruder summary is the more robust one.

The out-of-support detector of section 6.4 never fired.

8. Discussion

We separate the rules we expect to survive outside this environment from the ones that may be artifacts of it.

Likely to generalize.

1. *An acquired prior must never overrule present evidence.* The one collapse we observed was entirely this: an atom observed solving a task in the world at hand was assigned zero probability because training had not seen it. Score every hypothesis at least as high as present evidence alone would.
2. *Hedge every acquired prior.* Reserving a share of the budget for the uninformed rule bounds the loss where the prior does not apply, needs no detector, and cost nothing in support in our measurements.
3. *Generalization reaches exactly as far as the representation it is stated over.* An identity prior cannot reach an unseen atom by construction. The evaluator-side corollary is that a holdout must cross a boundary the representation can see, or its failure is a property of the labelling.
4. *When a prior applies matters as much as how good it is.* The same experience was worth +86 [-40, 222] or +621 [384, 841] on when it could be applied.
5. *Charge the experience, or you are measuring a different quantity.* The value of accumulated experience and improvement productivity came apart by the whole of the effect.

Methodological, and we would apply them anywhere.

6. The adversarial default must include model capacity.
7. Read a gate across the whole resource range, and check its threshold is attainable before running it.
8. The free rule in an arm's class must be the best free rule available.
9. Tie-breaks are part of an arm.
10. Do not read a threshold off two points.

Possibly artifacts of this environment. That a cruder estimator beats a better-specified one; that outcome feedback is worth nothing in the earned loop; the precise shape of the ceiling-recovery curve.

9. Limitations

- One environment, one task form, one alphabet, one hypothesis language. Every rule above is a conjecture outside it.
- The hypothesis space is finite and enumerable by construction. This is what makes budget countable and the oracle exact, and it is the assumption most worth attacking.
- Grading reads a predicate as the longest satisfying prefix while the interpreter also accepts a later first run. Sound and 0.14 to 0.54 percent conservative, uniformly across arms.
- No forecast was committed before any run. The programme this belongs to asks for proper-scored predictions and we have none.
- G1 is close to tautological and we say so.
- Two early screens ran on an earlier catalog build and do not import against the current module. They are marked superseded in the source; the load-bearing one was re-established on the current build.
- The cold-start comparison reserved evaluation mechanisms from training by index order rather than by the shuffled stream. The prior is a function of atom keys alone, so no answer can leak, but that run should be repeated before the number is quoted outside this draft.
- Nothing here is recursive. Compounding grows the estimator's input; nothing improves the estimator, and the earned agent runs one fixed rule for its whole life.
- Related work is unread. See section 2.

10. Conclusion and future work

An instrument in which unfamiliarity is drawn rather than labelled makes four questions separable that are routinely merged, and makes a negative result cost a minute rather than a day. On it, experience of a latent structure improves the solving of unfamiliar mechanisms from that structure, and experience of many structures improves the acquisition of one never seen. Both survive predeclared thresholds, two replications, structure controls and an independent audit. With experience charged to the agent that uses it, a self-improvement loop closes and needs no feedback, and the single largest lever we found is a prior that can be applied before the agent's first success.

Four things are open, in the order they block.

1. Make an acquired prior reliably useful at every resource level, or show it cannot be. Three gates failed with the same shape. Two hypotheses are testable in under an hour on this instrument: either the head of the ordering is decided by concentration rather than direction, measurable by comparing orderings rather than outcomes, or the fitted weights are accurate in aggregate and wrong about the few atoms that matter most at the head. The next study must fix its budget range by a rule stated and justified before it runs.
2. Make the earned loop's ceiling reachable. The gap between 0.292 earned and 0.925 informed is entirely the rate at which correct atoms arrive. Active choice of which hypothesis to submit, when a likely-wrong one would be more informative, is untouched.

3. A second task form in the same catalog, to separate growth in this environment from growth in this readout. This is the cheapest item and the one that most affects how far any of this generalizes.
4. Recursion. The instrument can ask whether the procedure that turns experience into a prior can itself be improved from that experience. We have not asked it.

Appendix A: reproducibility

Everything runs on one workstation in Python 3.12 with numpy and scikit-learn. The environment and harness are two modules; each experiment is one script writing one JSON analysis file that refuses to overwrite itself. Thirteen tests cover the load-bearing invariant, which is that the bitmask grading agrees with the host interpreter candidate by candidate.

step	cost
catalog build, 166,596 rows	0.2 s, 65 MB
a development screen	15 to 90 s
confirmed run, two replications	220 s wall, 254 s CPU, 192 MB
independent audit, 31,336 interpreter checks	238 s
the whole sequence in this paper	under one hour of compute

Artifact inventory. Environment and harness: `rsi/mechanism_space.py`, `rsi/mechanism_worlds.py`, `tests/test_mechanism_space.py`. Screens and development runs: `mechanism_selection_recheck.py`, `mechanism_world_growth.py`, `mechanism_meta_growth.py`, `mechanism_meta_support.py`, `mechanism_meta_transfer.py`, `mechanism_family_transfer.py`, `mechanism_metric_transfer.py`, `mechanism_compounding.py`, `mechanism_lifelong.py`, `mechanism_takeoff.py`, `mechanism_cold_start.py`, `mechanism_hedged_start.py`, `mechanism_world_retrieval.py`, `mechanism_atom_clustering.py`, `mechanism_metric_worlds.py`, `mechanism_metric_form.py`, `mechanism_metric_identifiability.py`. Confirmations: `EXPERIMENT-mechanism-growth-confirmation.md` with `mechanism_growth_confirmation.py` and `audit_mechanism_growth_confirmation.py`; `EXPERIMENT-metric-world-transfer-confirmation{,-2,-3}.md` with `metric_world_confirmation{,2,3}.py`. Results record: `MECHANISM-GROWTH-RESULTS-2026-09-21.md`. Prior programme: `WHAT-WORKED-WHAT-FAILED-2026-09-21.md`.